CMSC 320 Final Project — Checkpoint 2

S&P 500 Stock Market EDA

This notebook preprocesses the S&P 500 dataset and explores it with three statistical analyses: descriptive statistics on daily returns, a one-way ANOVA across sectors, and a t-test comparing high vs low volume days.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import f_oneway, ttest_ind
```

I load three CSV files: daily stock prices, company metadata, and the aggregate index.

```
stocks    = pd.read_csv('/content/sp500_stocks.csv')
companies = pd.read_csv('/content/sp500_companies.csv')
index     = pd.read_csv('/content/sp500_index.csv')

print(stocks.shape, companies.shape, index.shape)
stocks.head()
```

```
(1891536, 8) (502, 16) (2517, 2)
```

	Date	Symbol	Adj Close	Close	High	Low	Open	Volume
0	2010-01-04	MMM	NaN	NaN	NaN	NaN	NaN	NaN
1	2010-01-05	MMM	NaN	NaN	NaN	NaN	NaN	NaN
2	2010-01-06	MMM	NaN	NaN	NaN	NaN	NaN	NaN
3	2010-01-07	MMM	NaN	NaN	NaN	NaN	NaN	NaN
4	2010-01-08	MMM	NaN	NaN	NaN	NaN	NaN	NaN

Next I parse dates, coerce numeric columns, merge sector labels, and compute daily returns.

```
stocks['Date'] = pd.to_datetime(stocks['Date'])
for col in ['Open', 'High', 'Low', 'Close', 'Adj Close', 'Volume']:
    stocks[col] = pd.to_numeric(stocks[col], errors='coerce')

stocks.columns    = stocks.columns.str.lower().str.replace(' ', '_')
companies.columns = companies.columns.str.lower().str.replace(' ', '_')

stocks = stocks.sort_values(['symbol', 'date']).reset_index(drop=True)
stocks = stocks.merge(companies[['symbol', 'sector']], on='symbol', how='left')
stocks['daily_return'] = stocks.groupby('symbol')['close'].pct_change()

df = stocks.dropna(subset=['daily_return', 'sector']).copy()
print(df.shape)
df.head()
```

```
/tmp/ipykernel_17367/198514948.py:10: FutureWarning: The default fill_method='ffill' in SeriesGroupBy.pct_change is d
  stocks['daily_return'] = stocks.groupby('symbol')['close'].pct_change()
(617659, 10)
```

	date	symbol	adj_close	close	high	low	open	volume	sector	daily_return
8291	2013-01-03	ABBV	21.450581	34.830002	35.000000	34.160000	35.000000	16739300.0	Healthcare	-0.008257
8292	2013-01-04	ABBV	21.179594	34.389999	34.889999	34.250000	34.619999	21372100.0	Healthcare	-0.012633
8293	2013-01-07	ABBV	21.222700	34.459999	35.450001	34.150002	34.150002	17897100.0	Healthcare	0.002035
8294	2013-01-08	ABBV	20.760809	33.709999	34.639999	33.360001	34.290001	17863300.0	Healthcare	-0.021764
8295	2013-01-09	ABBV	20.877821	33.900002	33.950001	33.610001	33.610001	18800400.0	Healthcare	0.005636

Overview of the clean dataset:

```
print('Rows:', df.shape[0], ' Columns:', df.shape[1])
print('Unique stocks:', df['symbol'].nunique())
print('Unique sectors:', df['sector'].nunique())
print('Date range:', df['date'].min().date(), 'to', df['date'].max().date())
```

```
Rows: 617659  Columns: 10
Unique stocks: 172
Unique sectors: 11
Date range: 2010-01-05 to 2024-12-20
```

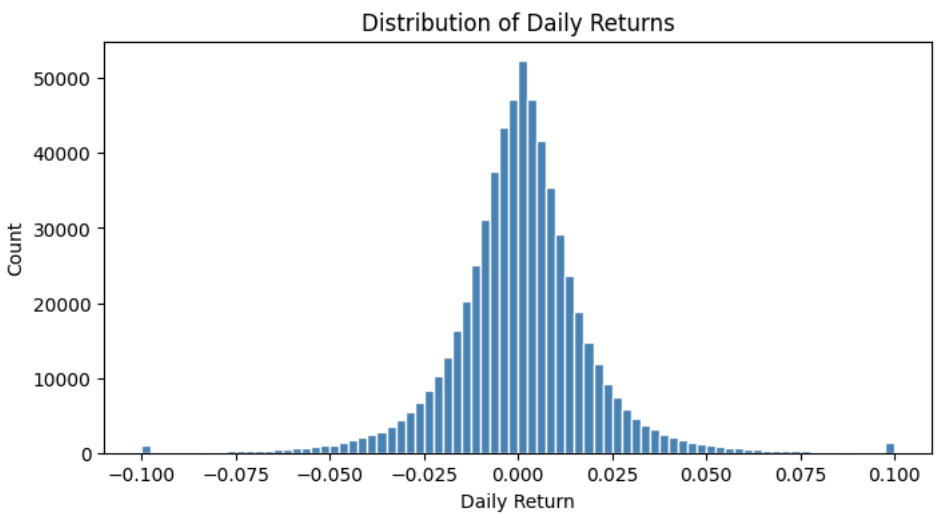
Analysis 1: Distribution of Daily Returns

I use descriptive statistics and a histogram to summarize the shape and spread of daily returns across all S&P 500 stocks.

```
returns = df['daily_return']
print(returns.describe())

plt.figure(figsize=(8, 4))
plt.hist(returns.clip(-0.1, 0.1), bins=80, color='steelblue', edgecolor='white')
plt.xlabel('Daily Return')
plt.ylabel('Count')
plt.title('Distribution of Daily Returns')
plt.show()
```

```
count    617659.000000
mean          0.000652
std           0.020048
min          -0.511559
25%          -0.008210
50%           0.000714
75%           0.009585
max           0.676812
Name: daily_return, dtype: float64
```



The mean daily return is close to zero, and the distribution is roughly bell-shaped, but there are still many extreme positive and negative return days. This tells me that daily returns are not perfectly normal.

Analysis 2: Do Sectors Have Different Mean Returns? (ANOVA)

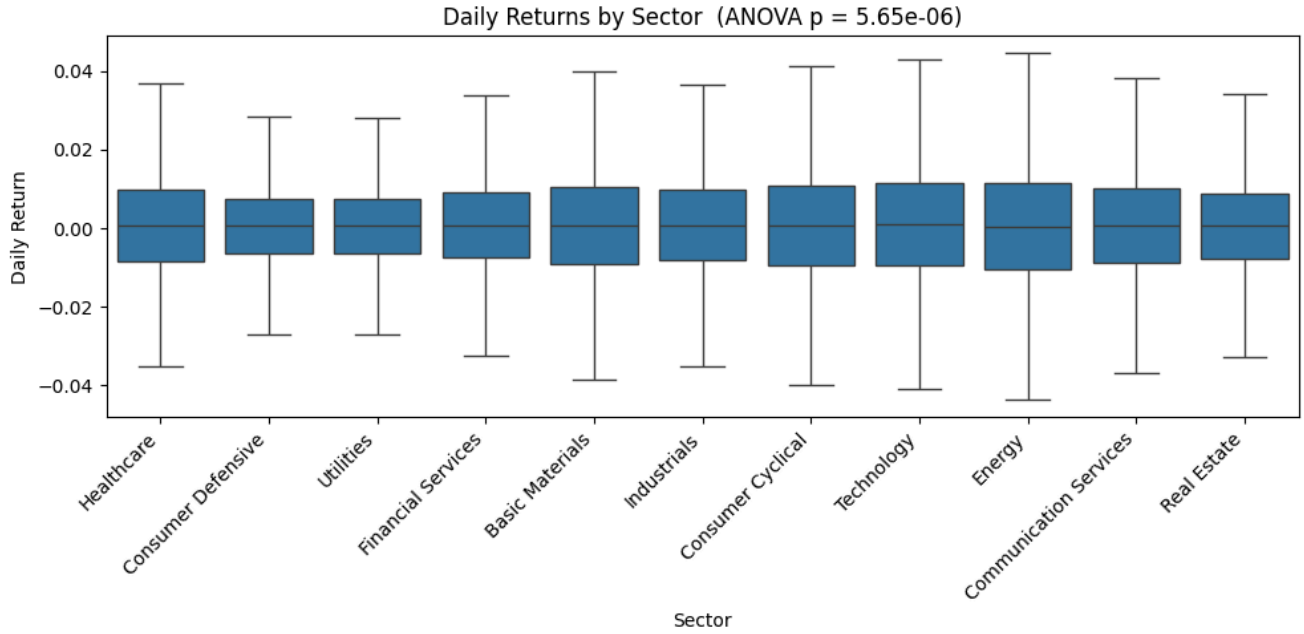
I run a one-way ANOVA to test whether mean daily returns differ significantly across the 11 S&P 500 GICS sectors.

```
groups = [g['daily_return'].values for _, g in df.groupby('sector')]
stat, p = f_oneway(*groups)
print('F-statistic:', round(stat, 4))
print('p-value:', p)

plt.figure(figsize=(10, 5))
sns.boxplot(data=df, x='sector', y='daily_return', showfliers=False)
plt.xticks(rotation=45, ha='right')
plt.xlabel('Sector')
plt.ylabel('Daily Return')
plt.title(f'Daily Returns by Sector  (ANOVA p = {p:.2e})')
```

```
plt.tight_layout()
plt.show()
```

```
F-statistic: 4.2691
p-value: 5.64937244087733e-06
```



Since the p-value is much smaller than 0.05, I reject the null hypothesis that all sectors have the same mean daily return. This suggests that average daily returns differ across sectors.

Analysis 3: High Volume vs Low Volume Days (t-test)

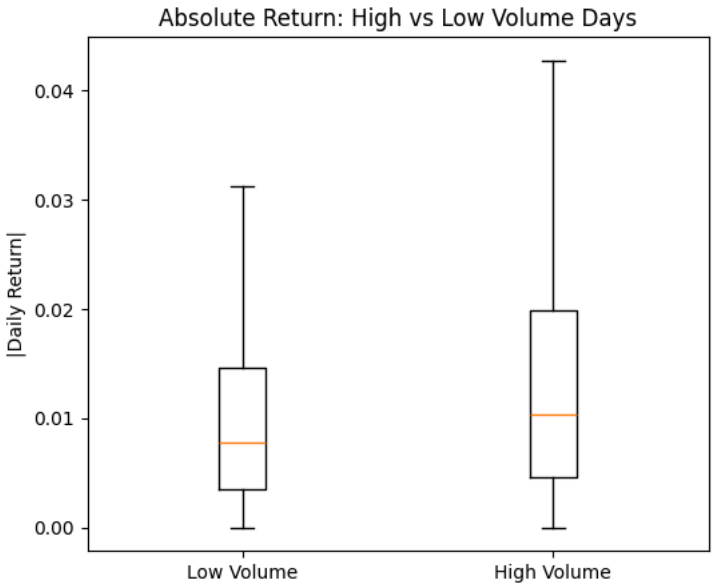
I split trading days at the median volume and use a two-sample t-test to check whether high-volume days tend to produce larger absolute returns than low-volume days.

```
median_vol = df['volume'].median()
high = df[df['volume'] >= median_vol]['daily_return'].abs()
low  = df[df['volume'] <  median_vol]['daily_return'].abs()

stat, p = ttest_ind(high, low, equal_var=False)
print(f"p-value: {p:.6e}")
print('high-vol mean:', round(high.mean(), 6))
print('low-vol mean: ', round(low.mean(), 6))

plt.figure(figsize=(6, 5))
plt.boxplot([low, high], tick_labels=['Low Volume', 'High Volume'], showfliers=False)
plt.ylabel('|Daily Return|')
plt.title('Absolute Return: High vs Low Volume Days')
plt.show()
```

```
p-value: 0.000000e+00
high-vol mean: 0.015355
low-vol mean:  0.01086
```



Since the p-value is much smaller than 0.05, I reject the null hypothesis that high-volume and low-volume days have the same average return magnitude. High-volume days tend to have larger price swings.